

Gain over a Judge-Free Ranker: Measuring a Typed-Decision Seam against a Strong Simple Baseline

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Abstract

A typed decision seam in an agent runtime routes each user turn to one of 115 skills whose rubrics are all shown to the judge, plus an option to decline. Because every candidate is exposed, an input-exposure control of the kind used to interpret grounded generation is vacuous here: the exposed fraction is 1, and gain over exposure is accuracy minus one, which for our 4B local engine is -0.116 . What this note measures instead is *gain over ranker* (g_{rank}), accuracy minus the accuracy of the best judge-free ranker over the same exposed options, with its decline threshold oracle-tuned on the same corpus. That is a different instrument with different guarantees, and the two are reported side by side so they cannot be confused. The control's first finding was a fault in itself: two defects in the baseline, a threshold sweep that could not reach its own optimum and exclusion clauses indexed as positive evidence, had manufactured 6.9 of an apparent 11.6-point advantage. Repaired, a BM25 ranker reaches 83.7% against the local engine's 88.4% ($g_{\text{rank}} = +4.7$, exact McNemar $p = 0.455$) and a cloud judge's 93.0% ($+9.3$, $p = 0.057$); against a semantic ranker at 69.8% both judges win decisively. Exact power at $n = 86$ against the observed lexical effect is 0.50. The corpus is self-authored and single-seed, and the tuning is oracle rather than held out, so these are not deployment estimates. The transferable findings are the repair discipline, the population breakdown, and the use of ranker-judge disagreement as a label-audit queue, which found a corpus error its authors had already reviewed.

1 What this note is

This is a companion note to a paper on exposure accounting for ontology-grounded generation over a curated corpus.¹ That paper's instrument classifies each gold item by whether the shown context exposed it and whether the answer recovered it, and summarises the table as a scalar: the recall a verbatim copy of the shown text would achieve, and the signed gain above it. The instrument is deterministic and judge-free, in the tradition of input-only baselines used to deflate reading-comprehension and inference benchmarks [2,

4, 6, 8] and of the broader argument that strong simple baselines should be run before elaborate pipelines are believed [5].

Applying that instrument to a second, differently shaped task exposed a boundary in it. On a discriminative choice over a curated option set, every candidate is shown by construction, so the exposed fraction is 1 and the exposure-based gain degenerates to accuracy minus one. Something else has to play the role of the copy baseline, and what we built is a judge-free ranker over the same exposed options. That is a legitimate and, we argue, a necessary comparison for a seam of this kind. It is not the exposure scalar, and the substitution was the part the original section got wrong. This note states both quantities, reports the routing measurement against the correct one, and keeps the material that survives contact with the corrections: two self-inflicted defects, three engines, a population breakdown, a label audit and the reasons none of it is a deployment claim.

2 The seam and its total exposure

An agent runtime routes each user turn to one of 115 *skills*. Each skill carries a short rubric stating what it is for and, in 50 of the 115 cases, what it must not be chosen for and which skill to use instead. The router puts one typed choice to a judge: the turn as state, the 115 rubrics as options, and a **none** option for turns no skill should handle. The gold label for a turn is the key of one exposed rubric, or **none**.

Exposure is therefore total. The gold answer is, by construction, among the text the judge is shown, which is stronger than the generation study the parent paper reports, where 3.6% of gold lay outside the retrieved scaffold. Total exposure is exactly the condition under which an exposure control stops discriminating: it cannot separate a judge that reads the rubrics from one that picks at random from the same shown set.

We wrote the corpus: 86 labelled turns (16 **none**, 70 skill), each tagged by the class of confusion it was written to probe. The corpus, the three engines and the measurement rig are released with the runtime.²

¹The *Copy Ceiling: An Input-Exposure Control for Ontology-Grounded Generation over Curated Corpora*, DreamLab AI, 2026. The routing study reported here formed a section of that paper's seventh draft and was separated from it on 21 September 2026 following external review, because it reports a different quantity under a name the paper had already spent.

²Sovereign System One, in the agent runtime <https://github.com/DreamLab-AI/agentbox>: a typed-decision wire protocol (choice, score, probability) with a local engine behind it. The router's request is byte-identical across all three judges below, so the engine is a URL chosen by this measurement. Decision record: ADR-2095; rig:

Table 1: The two quantities on the same 86 turns. g_{exp} is the parent paper’s exposure scalar and is vacuous here because every option is shown ($e = 1$). g_{rank} is measured against the repaired lexical ranker ($r = 0.837$) and is the quantity this note reports.

Engine	a	g_{exp}	g_{rank}
421M encoder	0.535	−0.465	−0.302
4B cross-encoder	0.884	−0.116	+0.047
Cloud judge	0.930	−0.070	+0.093

3 Two quantities, named apart

Let a be the seam’s top-1 accuracy over the $N = 86$ turns, e the fraction of turns whose gold answer appears in the shown options, and r the top-1 accuracy of a specified judge-free ranking procedure over those same options.

$$g_{\text{exp}} = a - e \quad (\text{gain over exposure}) \quad (1)$$

$$g_{\text{rank}} = a - r \quad (\text{gain over ranker}) \quad (2)$$

On this seam $e = 1$ by construction, so g_{exp} is $a - 1$ and is negative for every engine measured. That is not a defect in the engines; it is the exposure control reporting, correctly, that it has nothing to say about a task where everything is exposed. Table 1 gives both numbers for the three engines so that neither can stand in for the other.

The distinction matters for a claim the original draft made and this note withdraws. That draft observed that one definition of the baseline, implemented three ways, moved the reported gain by about seven points while the classification of which rubric each turn’s gold names never moved, and read this as evidence that an exposure scalar is unstable where its underlying counts are not. It is not evidence of that. The counts stayed fixed because the shown options and the model’s answers stayed fixed. The reported number moved because we changed a different procedure, namely the ranker used as the baseline. What was demonstrated is **baseline sensitivity in g_{rank}** , which is a property of the instrument introduced here, not instability in the exposure scalar of the parent paper. The general lesson survives the correction and is worth more than the claim it replaces: when a task has no natural no-op extractor, the baseline has to be constructed, and every construction choice is somewhere the number can move.

4 Constructing the ranker baseline

A no-op extractor has no meaning when the answer is a key rather than a passage, so the baseline becomes the best judge-free surface match over the exposed options: rank the 115 rubrics against the turn and take the top.

`crates/system-one/system-one-eval`.

Table 2: The correction cascade. The engine’s answers and the corpus labels are identical in every row; only the baseline procedure changes. Counts are correct turns of 86, and g_{rank} is against the 4B cross-encoder at 76/86.

Lexical baseline	n	r	g_{rank}
41-point grid, all text	66	0.767	+0.116
+ breakpoint sweep	68	0.791	+0.093
+ clauses dropped	72	0.837	+0.047

We use two rankers with no trained judgement in them: BM25 over the rubric text ($k_1=1.5$, $b=0.75$, a fixed stoplist), and cosine similarity under a 384-dimensional sentence embedding.

A ranker has no rubric to match a decline against, so conceding **none** to the judge would hand the judge a large and spurious advantage. Each ranker is therefore given a decline rule whose threshold is tuned to maximise *its own* accuracy on the evaluated corpus. This is oracle tuning, and it is favourable to the baseline by design: within the chosen ranker family, on this corpus, no threshold does better. Section 9 states what that does and does not buy.

5 The control’s first finding was a fault in itself

Our first implementation reported a lexical baseline of 76.7% (66/86) and a judge advantage of +11.6 points. Both figures were artefacts of two defects in the baseline, each of which inflated the judge.

- The threshold sweep could not reach its own optimum.** A “decline below t ” rule is a step function whose only breakpoints are the observed scores, but the sweep evaluated a fixed 41-point grid spanning the observed range. The maximising window here is $9.5715 < t \leq 9.7420$, narrower than one grid step, and the sweep stepped over it: two **none** turns were recoverable at no cost to any correct answer and were never recovered. Enumerating every breakpoint moves the baseline to 79.1% (68/86) and g_{rank} to +9.3.
- Exclusion clauses were indexed as positive evidence.** Fifty of the 115 rubrics carry a clause saying what the option is *not* for and which option to use instead. Indexed in one undifferentiated bag, those clauses are dense in exactly the vocabulary they disclaim, so a turn about a topic scored *highest* on the rubric that exists to reject it. Dropping each clause from its marker onward moves the baseline to 83.7% (72/86) and g_{rank} to +4.7.

Together the two defects manufactured 6.9 of the original 11.6 points ($11.6 \rightarrow 9.3 \rightarrow 4.7$; the original draft misreported this difference as 8.1). In count terms the baseline moved from 66 to 68 to 72 correct of 86 while the engine stayed at 76; Table 2 shows the cascade.

The second correction carries a caveat the original draft did not state. We argued there that dropping exclusion clauses “cannot introduce supervision” because it uses strictly less of the exposed text than the original procedure. Using less text is not the same as using it neutrally. Choosing a task-aware preprocessing rule after inspecting results can adapt a baseline to a benchmark without adding a single token, and that claim is withdrawn. What we can defend is weaker and still worth having: the rule is stated in advance of the numbers it produces, it is uniform across all 115 rubrics, it is motivated by a mechanism (negative clauses are lexically dense in what they disclaim) rather than by which turns it fixes, and it is implemented once in the released rig. Treat it as a declared preprocessing choice, not as a neutral one, and re-declare it before any future corpus is scored.

Neither defect is a matter of taste. A baseline that indexes “NOT for X ” as evidence for X , or a tuning procedure that misses the maximum it claims to find, silently withdraws the guarantee that oracle-tuning the baseline exists to buy. An instrument meant to deflate other evaluations has to survive its own test, and ours did not until it was made to. Both defects are fixed in the released rig with regression tests that construct each failure.

6 Three engines against one repaired baseline

Table 3 reports three engines on the same 86 turns against the same two repaired baselines. They span a wide range of capability: a 421M-parameter encoder whose options share a single 256-token head budget and are each truncated to 48 tokens; a 4B-parameter natural-language-inference cross-encoder scoring each (turn, rubric) pair independently [10]; and a commercial cloud judge.

Against the embedding baseline both competent engines win decisively and the conclusion does not depend on which of the two baselines is chosen. Against the *lexical* baseline neither does: the cloud judge’s +9.3 points gives an exact p of 0.057 and the local cross-encoder’s +4.7 gives 0.455. The two are also not distinguishable from each other (93.0% against 88.4%; 8 discordant turns against 4, exact $p = 0.39$), so the table separates a judge from surface matching only in the semantic case, and does not separate the two judges from each other. Exact power at $n = 86$ against the observed lexical effect is 0.50; roughly 157 turns would be needed for 80%. Observed-effect power is a weak guide, and the operational conclusion is the pre-registration rule rather than the number: choose the smallest difference that would change an engine decision, and size the corpus for that before running it.

7 Where the difference lies, and where it does not

The aggregate hides the only part of this that a design choice predicted. Against the repaired lexical baseline the 4B engine’s advantage is *exactly zero* on ordinary near-neighbour picking (86.0% against 86.0%, $n = 50$) and exactly zero on early-discriminator turns (83.3% against 83.3%, $n = 54$), while all of it sits on boundary cases (+16.7, $n = 12$) and on turns whose discriminating clause falls late in its rubric (+12.5, $n = 32$). That is the correspondence worth reporting: total exposure predicts no gain on ordinary picking, and against a competent ranker one measures none.

No subgroup contrast survives Holm correction within its partition, and the two smallest carry four discordant pairs each, where the two-sided exact p floors at 0.125 and no effect size can reach significance. We report the pattern as a direction the design predicted, not as a set of established contrasts. The late-discriminator tag also depends on a 48-token cut inherited from a superseded engine, so it is a property of one configuration.

8 Disagreement as a label audit

The ranker and the engine err on different turns, and their union is right on 95.3% against 88.4% for the engine alone. Only 26 of the 86 turns can move a paired comparison at all: the 22 on which the procedures disagree and the 4 both get wrong. A relabelling inside the 60 turns both get right is exactly neutral, since a wrong gold there makes both wrong and leaves the discordant counts untouched. We therefore re-read the complete movable set against the rubrics’ own text, rather than only the turns favouring the ranker.

It found one corpus error. For “*Take a screenshot of the staging login page and fill in the demo credentials*” the corpus said **browser**, both judges said **browser-automation**, and the **browser** rubric itself reads “for a browser task choose **browser-automation**, not this”. The judges followed the rubric’s stated boundary; the ranker won on vocabulary overlap. The audit is symmetric by construction rather than by assertion, and the symmetry is testable: turn 28 is the mirror image, where the judge’s pick carries the clause “for a visual critique of an existing UI choose **design-audit**” and the same criterion points *away* from the judge, so the gold was kept; turn 18 is contestable in the judge’s favour and was left standing. Correcting the one label moves g_{rank} by 2.3 points and changes no conclusion here, which is the state a single contested label should be in.

A ranker wins by surface, a judge by the rubric’s semantics, so systematic disagreement between them localises either a mislabelled item or a genuinely ambiguous one. When gold derives from exposed text, the judge-free baseline is therefore also a pointer at the rows to re-adjudicate: disagreements go to an audit queue rather than being counted silently against either

Table 3: Typed skill routing, $n = 86$, g_{rank} against the repaired baselines. Both baseline columns are judge-free procedures over the same 115 exposed rubrics, each with an oracle-tuned decline rule. Exact two-sided McNemar on the discordant pairs [1]. Every figure is a single pass on one self-authored corpus.

Engine	Top-1	vs lexical baseline 83.7%		vs embedding baseline 69.8%	
		g_{rank}	Exact p	g_{rank}	Exact p
421M encoder, shared head budget	53.5	−30.2	—	−16.3	—
4B cross-encoder, independent scoring	88.4	+4.7	0.455	+18.6	0.00086
Cloud judge (jev-1.13.0)	93.0	+9.3	0.057	+23.3	< 0.001

procedure. This is the one claim here that needs no statistical power and that a stronger baseline cannot take away, and the only finding in this note that survived every correction above unchanged.

9 Caveats

- **Oracle tuning is not a held-out estimate.** The decline threshold is tuned to maximise the baseline’s accuracy on the same 86 turns it is then scored on. That makes the baseline favourable *within the chosen ranker family on this corpus*, which is the direction we want for a conservative reading of g_{rank} . It does not make g_{rank} a general lower bound on what a judge adds, and it is not a deployment number. Before g_{rank} is used to justify an engine in production it needs either a held-out tuning split or a resampling procedure that repeats the tuning inside each resample.
- **The exclusion-clause rule is a declared choice.** See §5. It is task-aware preprocessing selected after seeing results, and the claim that it cannot introduce supervision is withdrawn.
- **Benchmark leakage inflates any surface baseline.** The corpus is self-authored: 86 turns, single seed, prompts written by the same hands as the rubrics they must match. Mean content-token overlap between a turn and its own gold rubric is 37.1%, against 2.7% for a random rubric. The lexical baseline reported here is therefore high for a reason particular to this benchmark.
- **The cloud judge is non-deterministic.** It returned confidences of 0.95, 0.92 and 0.93 on identical repeated requests. Its column is a single pass and should carry repeats before any comparison is drawn between the two competent engines.
- **The three engines differ in training data as well as architecture**, so Table 3 is not a controlled architecture comparison. Judge idiosyncrasy and self-preference apply to the cloud column as they do to any model-as-judge result [7, 11].
- **Total exposure is not answer sufficiency.** Showing the gold rubric among 115 options is not the same as supplying the answer, and a competent choice among exposed options can require reading two clauses against each other. The exposure family

of instruments constrains what an evaluation may attribute; it does not establish what a model did internally [3, 9].

10 What was cut during preparation

Four analyses were dropped rather than reported, and the reasons are part of the record. Confidence-calibration, escalation-cascade and threshold-sweep tables were about the engine rather than the instrument, or were structurally incapable of significance at their n . A production reweighting was retracted on re-analysis: its interval was carried by two discordant turns in a 16-item stratum resampled to hold 65% of the estimand. The threshold-sweep robustness claim survives as one sentence rather than a table, because nine passes over the same 86 turns are one corpus re-scored nine times, the full 5.8-point span is five turns, and the widest pairwise contrast is itself non-significant. A cascade that ranks cheaply and escalates only on boundary and late-clause signals is suggested by §7 and is viable, but its figures were computed against the defective baseline and must be re-derived before it is built.

11 Reproduction

```
system-one-eval run --concurrency 1
--timeout-ms 120000, then system-one-eval
copy-ceiling --from-report ... --ceilings
both, against the released corpus at
tests/system-one/routing-cases.json. Three
operational notes cost this preparation time and
belong with the command. The local 4B engine is
serialised by a tokeniser lock, so the default con-
currency queues past the timeout. Embeddings must
be requested one text per call: the estate’s endpoint
returns global batch offsets in data[].index when
concurrent requests merge. A remote backend requires
a bearer token and the rig now refuses to run without
one, because an unauthenticated remote run answers
403 on every case and is otherwise indistinguishable in
the report from the backend being unavailable.
```

12 What this does and does not show

The instrument transferred to a discriminative task only by being replaced: where every option is exposed, an exposure control is vacuous and a judge-free ranker has to take its place. Under its own name, g_{rank} , what survives is narrow and worth stating exactly. Against a semantic ranker a competent judge wins decisively. Against a well-built lexical ranker on this corpus it does not win measurably, at power 0.50. The difference is located on boundary and late-clause turns and is exactly nil on ordinary picking. The baseline’s first two implementations manufactured two-thirds of the effect they were measuring, and the repaired control found a labelling error its authors had already reviewed.

The method claim is what generalises. A choice task whose gold is among its shown options should report a fair judge-free baseline beside its accuracy; should tune that baseline to a real optimum and declare every preprocessing rule that built it; should name the resulting quantity for the baseline it is measured against rather than borrowing the name of a different instrument; should read the gain by population rather than in aggregate; and should send the disagreements to an audit queue rather than counting them against either procedure. None of that requires a judge, a label budget or a model swap, which is why it is worth making standing practice at every typed-decision seam rather than an occasional exercise.

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